

Kai Xiao

Chinese

11 Avenue Roberto Rossellini
69100 Villeurbanne
France

Cell phone: +33 06 67 02 09 83
E-mail address: xiao@math.univ-lyon1.fr

Education

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| 2023 - now | PhD in mathematics , <i>Université Claude Bernard Lyon 1 – Institut Camille Jordan</i>

Thesis : <i>Sobolev spaces to manifolds: traces, singularities, weak density</i>
Thesis advisor : Professor Petru Mironescu |
| 2020 - 2023 | Master in mathematics , <i>Beijing Normal University, School of Mathematical Sciences</i>

Thesis : <i>Regularity of the Solution and Free Boundary of Porous Medium Equations on Riemannian Manifolds</i>
Thesis advisors : Professor Jingang Xiong |
| 2016 - 2020 | Bachelor in mathematics , <i>Beijing Normal University, School of Mathematical Sciences</i> |

Publications

- [1] Tristan Bullion-Gauthier and Kai Xiao, *L^2 -regularity of minimizers of anisotropic interaction functionals* (2026), submitted for publication, available at <https://hal.science/hal-05743076>.
- [2] Antoine Dettaille, Petru Mironescu, and Kai Xiao, *Pullback of closed forms by low regularity maps to manifolds, and applications*, Calc. Var. Partial Differential Equations **65** (2026), no. 198.

Awards and grants

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| 2022 | First Prize of Academic Scholarship of Beijing Normal University |
| 2022 | Excellent Teaching Assistant of School of Mathematical Sciences |
| 2021 | First Prize of Academic Scholarship of Beijing Normal University |

2020	First Prize of Freshmen Scholarship of Beijing Normal University
2019	First Prize of 'Jingshi' Scholarship of Beijing Normal University
2019	First Prize (Meritorious Winner) of MCM/ICM
2018	Second Prize of the Contemporary Undergraduate Mathematical Contest in Modeling
2018, 2017	Second Prize of 'Jingshi' Scholarship of Beijing Normal University

Talks

June 2026	Będlewo , Calculus of Variations in Material Sciences, <i>Sobolev maps to manifolds: the distributional Jacobian and its applications</i>
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Teaching experience

2022 - 2023	Teaching assistant of Graduate Course: Real Analysis , <i>Beijing Normal University</i>
2022	Teaching assistant of Graduate Course: Partial Differential Equations , <i>Beijing Normal University</i>

Languages

Mandarin	Native speaker
English	Upper intermediate (B2)
French	Utilisateur débutant (A2)

Leisure

Sport	Tennis, basketball
Miscellaneous	Video games, animation